FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Main)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST –II
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A	D
2.	A	B	B
3.	A	D	B
4.	D	C	B
5.	B	B	B
6.	C	D	B
7.	C	C	D
8.	B	B	C
9.	B	A	C
10.	B	D	C
11.	A	A	A
12.	C	A	D
13.	B	C	A
14.	B	C	B
15.	C	D	D
16.	B	C	B
17.	B	C	C
18.	B	D	B
19.	D	B	C
20.	C	A	C
21.	B	C	C
22.	C	D	D
23.	A	D	A
24.	A	C	A
25.	B	C	B
26.	B	D	C
27.	C	B	A
28.	A	C	B
29.	D	A	B
30.	B	C	C

Physics PART – I

SECTION - A

1. Q = C.V. , [Q] = T'A'
So, a + b + c + d + e + f + g + h = 2

3. $\vec{r}(t) = \vec{r}(0) + \int_{t=0}^{t=t} \vec{v}.dt$, $\vec{v}(t) = \int_{t=0}^{t=t} \vec{a}.dt + \vec{v}(0)$

$$\int_{t=0}^{t=t} \vec{v}.dt = \int_{0}^{t} \left[\left(t^2 + 5\right)\hat{i} + \left(\frac{3t^2}{2} + 2\right)\hat{j} \right] dt$$

$$= \left[\frac{t^3}{3} + 5t\right]\hat{i} + \left[\frac{t^3}{2} + 2t\right]\hat{j}$$

$$\vec{r}(t) = \left[\frac{t^3}{3} + 5t + 10\right]\hat{i} + \left[\frac{t^3}{2} + 2t + 10\right]\hat{j}$$

4. $$|\vec{a}_1| = |\vec{a}_2| = |\vec{a}_3| = \frac{T_A}{(m_1 + m_2 + m_3)} = \frac{T_B}{m_3 + m_2} = \frac{T_c}{m_3}$$

$\Rightarrow T_A > T_B > T_C$

$$T_A = T_B\left[1 + \frac{m_1}{m_2 + m_3}\right] \text{ and } \frac{m_1}{m_2 + m_3} < 1$$

So, $T_A < 2T_B$

$$T_B = T_C\left[1 + \frac{m_2}{m_3}\right] \text{ and } \frac{m_2}{m_3} < 1$$

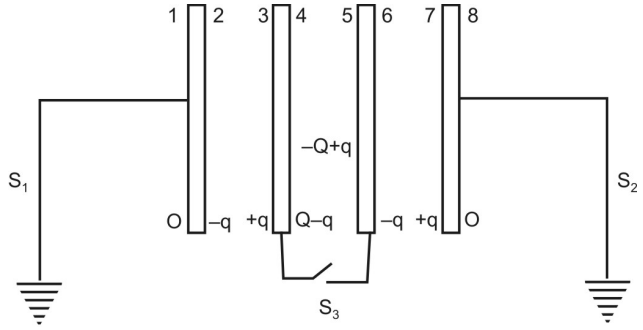
So, $T_B < 2T_C$

Now, $$T_A = T_C\left[1 + \frac{m_1 + m_2}{m_3}\right]$$

And, $m_1 + m_2 < m_3 + m_3 \Rightarrow \frac{m_1 + m_2}{m_3} < 2$

Or $T_A < 3T_C$

5. 'T' total charge in the system (initially) = zero We know that when two or more charged conducting plates are peaced parallel to each other, the outermost surfaces get equal charges 'p' can be proved using gauss's law) Where, $P = \frac{T}{2}$; So, P = 0



Since the middle plates are not connected total change on than would be conserved the charges on the plates after S_1 and S_2 are connected are shown in the figure. Again, we utilize the fact that the facing surfaces must carry opposite charges. Finally the potential difference between surface 1 and 8 must be zero. The electric field is non-zero between the conducting plates and zero inside them.

$$d\left(\frac{q}{A\varepsilon_0}\right)-\left(\frac{Q-q}{A\varepsilon_0}\right)d+d\left(\frac{q}{A\varepsilon_0}\right)=0$$

d = spacing between the plates A = Area of the plate

$\Rightarrow q = Q/3$

6. $x = A\cos wt$

$R = \Delta \sin wt : Q = A\cos wt = A\sin\left(\frac{\pi}{2}-wt\right)$

7. Electric field at a distance x-on the axis
= Electric field due to an infinite sheet
– Electric field due to a uniformly charged disc of radius 'R'

$$=\frac{\sigma}{2\varepsilon_0}-\frac{\sigma}{2\varepsilon_0}\left[1-\frac{x}{\sqrt{x^2+r^2}}\right]$$

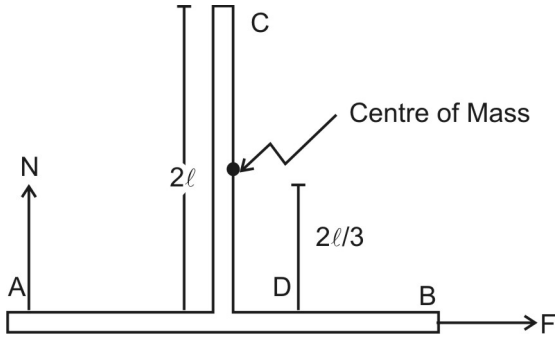
$$E=\frac{\sigma}{2\varepsilon_0}\frac{x}{\sqrt{x^2+R^2}}$$

$$F=qE=-\frac{\sigma ex}{2\varepsilon_0\sqrt{x^2+R^2}}$$

$$\Rightarrow mu\frac{du}{dx}=-\frac{\sigma ex}{2\in_0\sqrt{x^2+R^2}}$$

$$u=\sqrt{\frac{\left(\sqrt{2}-1\right)\sigma eR}{m\varepsilon_0}}$$

8. When the point B is about to leave ground the normal reaction to the rigid rod passes through A. Gravity would not cause a torque about centre of mass. Also, the torques must balance just before B leaves ground.



$$F\left(\frac{2\ell}{3}\right)=3mg\left(\frac{\ell}{2}\right)$$

$$F=\frac{9mg}{4}$$

9. $\vec{V}_{cm}$ of the system is non-zero initially and would remain so because $\vec{F}_{ext}=0$ at all times.

10. Initial angular momentum = mvr

Final angular momentum = $\left(\frac{mr^2}{2}\right)w + m(wr)r$

$\because$ T_{ext} = 0 : L is conserved

Mur = $\frac{3}{2}mwr^2$

Or, $wr = \frac{2u}{3}$

Velocity of mass m first after the string is taut = wr = 2m/s

11. Force on inclined surface in liquid
= Pressure at the centroid of the surface × area of surface

$= \left[\rho g\left(l + \frac{a}{2}\right)\sin\theta\right]ab$

12. $P_1 = \frac{1}{2}\rho u^2 + P_2$

$10^5 Pa = \frac{1}{2}\times 10^3 \times (10)^2 + P^2$

$P_2 = \frac{1}{2}\times 10^5 Pa$

13. $\gamma = \frac{P - \mu dg}{2d}$

P = 5mN, μ = 1.75×10^{-3} kg/m, d = 50 mn

14. Potential energy = – 2 × kinetic energy
Magnitude of potential energy = 2 × K.E.
Clearly, U > K

16. Volume of the gas is increasing (assuming its mass remains constant) and volume of the gas is directly proportional to T.

nC_v (dT) = (dV) $\frac{k}{m}$ or V$\propto$ T

The process is isobaric and exchange of heat is given by nC_p (dT)

17. $\lambda_3 = \frac{2\ell}{3} = \frac{1}{3}m$

Wave speed on the string = $\nu\lambda_1$ (lowest resonant node) u_{string} = 400 m/s
When a wave passes from one medium to another (the air) of differing wave speed, the frequency remains the same.

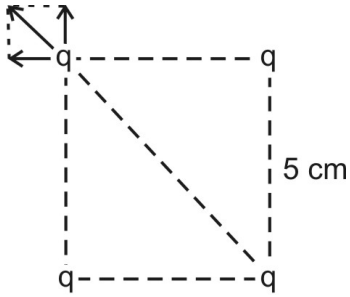
Now, $\lambda_{air} = \lambda_{string}\frac{v_{air}}{v_{string}}$

$= \frac{343}{400}\times\frac{1}{3}m$

18. $F_q = kq^2\left[\frac{\sqrt{2}}{\left(5\times10^{-2}\right)^2}+\frac{1}{\left(5\sqrt{5}\times10^{-2}\right)}\right]$

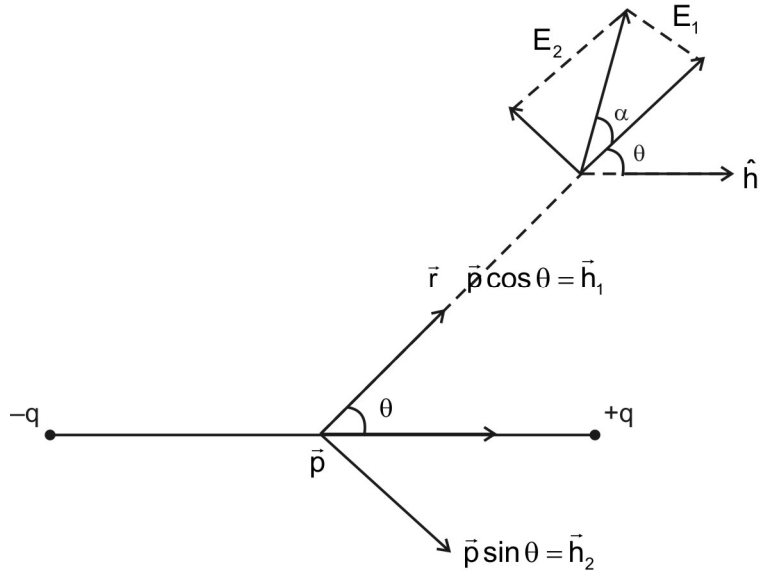
$=\frac{9\times10^9\times4\times10^{-12}}{25\times10^{-4}}\left[\sqrt{2}+\frac{1}{2}\right]N$

$= 27.5\ N$



20. Distance travelled in first two seconds = 15 m
Acceleration of the Car at t = 4s is zero
Braking force = Ma
= 1 Mg×15 m/s^2
= 15 KN

21. $E_1=\frac{2k(p\cos\theta)}{|\vec{r}|^3};E_2=\frac{k(p\sin\theta)}{|\vec{r}|^3}$



$\frac{E_2}{E_1}=\tan\alpha=\frac{\tan\theta}{2}$

or $\alpha=\tan^{-1}\left\{\frac{\tan\theta}{2}\right\}$

or $\psi=\theta+\tan^{-1}\left\{\frac{\tan\theta}{2}\right\}$

25. $\frac{P_1}{P_2}=\frac{H_1}{H_2}=\frac{\left(V^2/R_1\right)}{\left(V^2/R_2\right)}=\frac{R_2}{R_1}=\frac{2R}{R}=\frac{2}{1}$

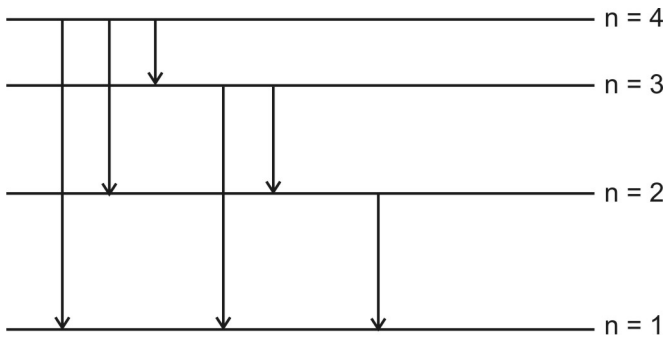
26. $C_{AB} = C/2$; $C_{CD} = 2C$; $C_{EF} = C$

27. A slit would have produced a spherical wave front. A biprism and a glass slab would produce a parallel wave front.

28. $\frac{1}{f_1} = (1.5-1)\left\{\frac{1}{25}\right) = \frac{1}{50};\ \frac{1}{f_2} = (1.8-1)\left\{\frac{1}{24}\right) = \frac{1}{30}$

$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{8}{150} = \frac{4}{75}; F = \frac{75}{4}\text{cm}$

29. $E = \frac{hc}{\lambda} = 12.79\text{eV}$



$E_n = E_1 + E = -0.81\text{eV}$

$E_n = -\frac{13.6}{n^2} = -0.81\text{eV}; n \simeq 4$

Chemistry

PART – II

SECTION – A

1. The mass of 1 cc of $(C_2H_5)_4Pb$ is = $1 \times 1.66 = 1.66$ g and this is the amount needed per litre.
No. of moles of $(C_2H_5)_4Pb$ needed = $\frac{1.66}{323}$ = 0.00514 ml
1 mole of $(C_2H_5)_4Pb$ requires $4 \times (0.00514) = 0.0206$ ml of C_2H_5Cl
$\therefore$ Mass of C_2H_5 Cl = $0.0206 \times 64.5 = 1.33$ g

2. The sp^3 bound leaving groups can be directly displaced by the nucleophile or the leaving group can ionise off to form a carbocation (S_N1) >C = C – L. However, sp^2 bound leaving groups are very difficult to ionise Halogen attached to allylic system are activated towards S_N1 due to stability of allyl carbocation.

4. Aromatic stabilisation occurs in rings that have an unbroken loop of p-orbitals. Any cyclic compound will be stable especially when the ring contains (4n + 2) π-electron.

5. The alcohols II & III are 3° but III gives more substituted C = C, I & IV are both 2° but IV can give a more substituted C = C.

6. $A_2X_3 \rightleftharpoons 2A^{3+} + 3X^{2-}$
$\quad\quad\quad

10. $N_2H_5^+ + H_2O \rightleftharpoons N_2H_4 + H_3O^+$

$K_h = \frac{[N_2H_4][H_3O^+]}{[N_2H_5^+]} = \frac{k_w}{k_b} = \frac{10^{-14}}{9.6\times10^{-7}} = \frac{x^2}{0.1} = 1.04 \times 10^{-8}$

$\therefore x = 3.2 \times 10^{-5}$

% hydrolysis = $\frac{3.2\times10^{-5}}{0.1}\times100 = 0.032\ \%$

11. $NH_3 + H^+ Cl \longrightarrow NH_4^+ Cl$

Solution contains NH_4Cl in contact with NH_3 (excess) & hence buffer

12. $2.303 \log C/C_0 = -kt$

40% to completion means 60% unreacted

$2.303 \log 0.6 = -(3.3 \times 10^{-4})t$

$\therefore t = 26$ min

13. 9.8 g Cu = $\frac{9.8}{63.5}\times2 = 0.308$ mol e^-

$10000 \times 3 \times \frac{1}{96500} = 0.311$ mol e^-

If 0.311 mol electrons provided by the current 0.308 mol was used to deposit copper.

The current efficiency is $\frac{0.308}{0.311}\times100 = 99.1\%$

14. Moles escaped = original – final moles

$= \frac{4\times10}{0.082\times290} - \frac{2.5\times10}{0.082\times290} = 0.630$ mol

20. $\frac{1.26\times10^{-4}}{(1.26\times10^{-4})+(3.8\times10^{-5})} = 77\%$ cyclohexene

$\therefore$ methyl cyclopentene = (100 – 77) = 23%

21. Let x and y be the molar mass of A & B respectively then

$\frac{\Delta T_f}{K_f} = m = 1 = \frac{8\times1000}{(x+2y)\times100} = \frac{80}{x+2y}$ … (1)

& $\frac{\Delta T_f}{K_f} = m = 1 = \frac{10\times1000}{(2x+y)\times100} = \frac{100}{2x+y}$ … (2)

Solving (1) & (2) x = 40; y = 20.

22. Reaction of D-(+)-glucose with methanolic HCl leads to the formation of methyl glucoside (C, - OH group is methylated) which, being acetal is not hydrolysable by base, so it will not respond Tollen's reagent.

23. For precise fitting of Zn^{2+} ions into tetrahedral voids of S^{2-} ions packing r^+ / r^- ratio must be 0.225.

24. $\bar{\nu} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

For last line of Lyman series of H-specturm; Z = 1, n_1 = 1, $n_2 = \infty$.
For Hα line in Balmer series of He^+; Z = 2, n_1 = 2, n_2 = 3.

25. NO^+ has lost an antibonding electron where as CO^+ has lost bonding electron.

26. Below CMC no micellization takes place. Sodium oleate ionizes almost completely in aqueous solution.

27. $\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \longrightarrow HCl(g)$ ... (1)

$\Delta S^\circ = \sum S^\circ(\text{product}) - \sum S^\circ(\text{reactants})$

$= S^\circ_{HCl} - \left[\frac{1}{2}S^\circ_{H_2} + \frac{1}{2}S^\circ_{Cl_2}\right]$

$= 187 - \frac{1}{2}[131 + 223] = 10\ JK^{-1}$

28. In PO_4^{3-}, P shows sp^3 hybridization.
In NO_3^-, N shows sp^2 hybridization'
In ICl_2^+, I shows sp^3 hybridization.

29. $OF_2 \rightarrow$ Pale yellow.
SF_6 & $SF_4 \rightarrow$ Gas.

Mathematics

PART – III

SECTION – A

1. Bisectors of the lines in both the cases will remain unchanged

2. $D_1 = D_0^2$, $D_2 = D_1^2 = D_0^4$, ... $D_n = D_0^{2^n}$

3. $f(x) = \left[\frac{-t}{\pi}\cos(x+\pi t)\right]_0^1 + \frac{1}{\pi}\int_0^1 \cos(x+\pi t)\,dt$

$= \frac{1}{\pi}\cos x - \frac{2}{\pi^2}\sin x$

$f_{max} = \sqrt{\frac{1}{\pi^2}+\frac{4}{\pi^4}} = \frac{\sqrt{\pi^2+4}}{\pi^2}$

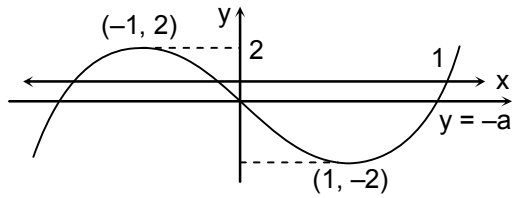
4. $y = x^3 - 3x$

$y = -a$

$\frac{dy}{dx} = 3x^2 - 3$

$\Rightarrow x = 1$ or -1

For three different roots $-a \in (-2, 2)$



5. $x = -i\cot\frac{k\pi}{n}$, $k \in \{1, 2, 3, \ldots, n-1\}$

Then $\frac{x}{1} = \frac{\cos k\frac{\pi}{n}}{i\sin\frac{k\pi}{n}}$

$\Rightarrow \left(\frac{x+1}{x-1}\right) = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}$

$\Rightarrow \left(\frac{x+1}{x-1}\right)^n = \cos 2k\pi + i\sin 2k\pi = 1$

$\Rightarrow (x+1)^n - (x-1)^n = 0$

$\Rightarrow {}^nC_1x^{n-1} + {}^nC_3x^{n-3} + \ldots = 0$

Roots of above equation are $-i\cot\frac{k\pi}{n}$, $k \in \{1, \ldots n-1\}$

$\therefore \sum_{k=1}^{n-1}\left(-i\cot\frac{k\pi}{n}\right)^2 = \left(\sum_{k=1}^{n-1}\left(-i\cot\frac{k\pi}{n}\right)\right)^2 - 2\cdot\sum\sum_{1\le p<q\le n-1}\left(-i\cot\frac{p\pi}{n}\right)\times\left(-i\cot\frac{q\pi}{n}\right) = 0 - 2\frac{{}^nC_3}{{}^nC_1}$

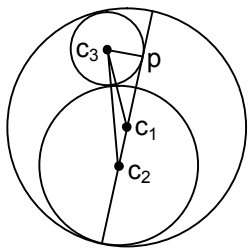
$\Rightarrow f(n) = \frac{(n-1)(n-2)}{3}$

6. $c_1 : \left(\frac{-5}{\sqrt{2}}, \frac{-5}{\sqrt{2}}\right)$

$r_1 = 5$

$c_2 : \left(\frac{-3}{\sqrt{2}}, \frac{-3}{\sqrt{2}}\right)$

$r_2 = 3$



r be radius of third circle and its centre be at c_3, then

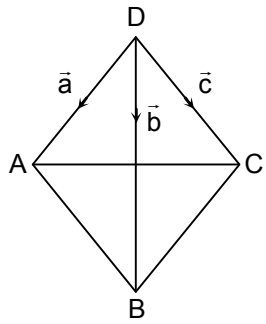
$$(c_2c_3)^2 = (pc_3)^2 + \left(c_1c_2 + \sqrt{\left((c_1c_3)^2 - (pc_3)^2\right)}\right)^2$$

$$\Rightarrow (r_2 + r)^2 = r^2 + \left(r_1 - r_2 + \sqrt{(r_1 - r)^2 - r^2}\right)^2$$

$$\Rightarrow r = \frac{4r_1r_2(r_1 - r_2)}{(r_1 + r_2)^2} = \frac{15}{8}$$

$\Rightarrow \lambda = 8$

7. D be the origin and $\overline{DA} = \bar{a}$, $\overline{DB} = \bar{b}$ and $\overline{DC} = \bar{c}$



$|\bar{a} - \bar{b}| = 41$, $|\bar{b} - \bar{c}| = 36$, $|\bar{c} - \bar{a}| = 7$

distance between mid-points of AB and CD

$$d^2 = \left(\frac{\bar{a} + \bar{b} - \bar{c}}{2}\right)^2 = \frac{1}{4}\left(|\bar{a}|^2 + |\bar{b}|^2 + |\bar{c}|^2 + 2\bar{a}\cdot\bar{b} - 2\bar{a}\cdot\bar{c} - 2\bar{b}\cdot\bar{c}\right)$$

$$= \frac{1}{4}\left(|a|^2 + |b|^2 - |c|^2 - |\bar{a} - \bar{b}|^2 + |\bar{a} - \bar{c}|^2 + |\bar{b} - \bar{c}|^2\right)$$

= 137

8. $2b = a + c \Rightarrow 2\sin B = \sin A + \sin C$

$$\Rightarrow 2\left(2\sin\frac{B}{2}\cdot\cos\frac{B}{2}\right) = 2\sin\frac{A + C}{2}\cdot\cos\frac{A - C}{2}$$

$$\Rightarrow \sin\frac{B}{2} = \frac{1}{2\sqrt{2}},\ \cos\frac{B}{2} = \frac{\sqrt{7}}{2\sqrt{2}} \Rightarrow \sin B = \frac{\sqrt{7}}{4}$$

9. Obviously for $c \in (0, 1)$, f(x) lies above the g(x), also $x^2 = cx^3$

$\Rightarrow x = 0,\ x = \frac{1}{c}$

Hence, $\int_0^{1/c}\left(x^2 - cx^3\right)dx = \frac{1}{12c^3} = \frac{2}{3}$

$$\Rightarrow c = \frac{1}{2}$$

10. sec 40º, sec 80º, sec 160º are the roots of $\frac{8}{t^3} - \frac{6}{t} + 1 = 0$

or $t^3 - 6t^2 + 8 = 0$

Sum of roots = 6

11. The chord of contact from (h, k) to $y^2 = 4x$ is $ky = 2(x + h)$. If it touches the hyperbola

$\Rightarrow 4h^2 + k^2 = 4$ locus is $\frac{x^2}{1} + \frac{y^2}{4} = 1$

12. $x = \sin^{-1}(a^4 + 1) + \cos^{-1}(a^4 + 1) - \tan^{-1}(a^4 + 1)$ is defined at a = 0 only

$$\Rightarrow x = \frac{\pi}{4}$$

$$\therefore \text{Area } A = \frac{1}{2}\cdot\frac{\pi}{4}\cdot\frac{\pi}{4} = \frac{\pi^2}{32}$$

13. This can happen, if three lines are real and distinct as well as angle between any two adjacent sides is $\frac{2\pi}{3}$

$\Rightarrow f(m) = bm^3 + dm^2 + cm + a = 0$ has three distinct real roots (where m = y/x)

And $\frac{m_1 - m_2}{1 + m_1 m_2} = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{m_3 - m_1}{1 + m_1 m_3} = \pm\sqrt{3}$

$\Rightarrow 3 + m_1m_2 + m_2m_3 + m_3m_1 = 0$ and $3bm^2 + 2dm + c = 0$ has roots α, β with $f(\alpha)\, f(\beta) < 0$

$\Rightarrow 3b + c = 0$

14. $2n\{x\} = 3x + 2[x]$

$\Rightarrow \{x\} = \frac{5[x]}{2n-3}$ as $0 \le \{x\} < 1$

$\Rightarrow 0 \le [x] \le \frac{2n-3}{5}$ $\{n \ge 2\}$

It has 5 solution, [x] = 0, 1, 2, 3, 4, only if $4 < \frac{2n-3}{5} \le 5$

$\Rightarrow \frac{23}{2} < n \le 14 \Rightarrow 12, 13, 14$

15. Let a is first term and d is common difference of the A.P.
$a + (r - 1)d = 6$, $a + (s - 1)d = 8$, $a + (t - 1)d = 12$
Solving, we get $2r + t = 3s$ also $f(x) = tx^2 + 2rx - 2s$
$\Rightarrow f(0) = -2s$, $f(1) = s$
$\Rightarrow f(0)\, f(1) = -2s^2 < 0$ hence, exactly one root in (0, 1)

16. $(a - b)^2 + (b - c)^2 + (c - d)^2 + (d - a)^2 \ge 0$
$\Rightarrow 2(a^2 + b^2 + c^2 + d^2) \ge 2(ab + bc + cd + ad)$
$\Rightarrow ab + bc + cd + ad \le 25$

17. f(x) is continuous where $3 \sin x + a^2 - 10a + 30 = 4 \cos x$
or $a^2 - 10a + 30 = 4 \cos x - 3 \sin x$
LHS ≥ 5, RHS ≤ 5
$\Rightarrow (a - 5)^2 + 5 = 4 \cos x - 3 \sin x$
$\Rightarrow a = 5$, $x = 2n\pi - \tan^{-1}\left(\frac{3}{4}\right)$, $n \in I$

18. Required ways = ${}^6C_2 \cdot {}^4C_2 = 90$

19. $z = x + iy$

$\Rightarrow \sqrt{(x-2)^2 + (y+1)^2} = \frac{1}{\sqrt{2}}|x - y|$

20. Let $\vec{a} = 3\hat{i} + 4\hat{j} + 5\hat{k}$

$\vec{b} = \tan x\hat{i} + \tan y\hat{j} + \tan z\hat{k}$

$(\vec{a} \cdot \vec{b})^2 \le |\vec{a}|^2 |\vec{b}|^2$

$\Rightarrow \tan^2 x + \tan^2 y + \tan^2 z \ge 8$

21. $P = \cos\theta\cos 2\theta \cos 3\theta \ldots\ldots \cos 1004\theta$
$Q = \sin\theta \sin 2\theta \ldots\ldots \sin 1004\theta$
Then 2^{1004} PQ = $\sin 2\theta \sin 4\theta \ldots\ldots \sin 2008\theta$
= $(\sin 2\theta \ldots\ldots \sin 1004\theta)[\sin(2\pi - 1003\theta)\sin(2\pi - 1001\theta) \ldots\ldots \sin(2\pi - \theta)]$
= $(\sin 2\theta \sin 4\theta \ldots\ldots \sin 1004\theta)(-\sin 1003\theta) \ldots\ldots (-\sin\theta) = Q$
$\Rightarrow p = \dfrac{1}{2^{1004}}$

22. $1 + x^2 = t^2 \Rightarrow x\,dx = t\,dt$
$I = \int \dfrac{tdt}{\sqrt{t^2 + t^3}} = \int \dfrac{dt}{\sqrt{1+t}} = 2\sqrt{1+t} + c = 2\sqrt{1+\sqrt{1+x^2}} + c$

23. If $OP \perp OQ$ for P and Q on the hyperbola then b > a
Director circle of the hyperbola is $x^2 + y^2 = a^2 - b^2$, that exist only when a > b

24. Any plane through first line can be written as $3x + y + 2z - 2 + \lambda(2x + y + z - 1) = 0$
It is parallel to x = y = z if $\lambda = -\dfrac{3}{2}$
Thus, plane parallel to 2nd line is $-y + z - 1 = 0$

25. $A^* = \left(\overline{A^+}\right)$. If $AA^* = I$, then A is unitary

26. Solution is y = m(x + 2)
Fixed point p is (–2, 0)
Area = $\dfrac{2}{\sqrt{3}}$

27. $\left(\dfrac{1}{x^2} + \dfrac{1}{x} + 1 + x + x^2\right)^5 = \dfrac{1}{x^{10}}\left(1 - x^5\right)^5 (1 - x)^{-5}$
Now, coefficient of x^{10} in $(1 - x^5)^5 (1 - x)^{-5}$ is = ${}^{14}C_4 - {}^5C_1 \times {}^9C_4 + {}^5C_2 \times {}^4C_4 = 381$

28. Mean = 1007
Variance $\sigma^2 = \dfrac{1}{1007}\left(1^2 + 3^2 + \ldots.. + 2013^2\right) - 1007^2$ = 338016
$\therefore\ d = \sigma = \sqrt{338016} = 581.39$

29. Let n(A) = n
Then $2^{n^2 - n} = 2^{\frac{n^2+n}{2}}$
$\Rightarrow n = 3$

30. $(\sin^{-1} x + \sin^{-1} y)(\sin^{-1} z + \sin^{-1} w) = \pi^2$ is possible only when x = y = z = w = 1 or –1 then
$\begin{vmatrix} x^{n_1} & y^{n_2} \\ z^{n_3} & w^{n_2} \end{vmatrix}$ = 0, 2 or –2